QUESTIONS

1. Explain each of these terms:
   1 Diversity
   2 Classify
   3 Species
   4 Heredity
   5 Gene
   6 DNA
   7 Complementary base pair
   8 Double helix
   9 Coding DNA
   10 Junk DNA
   11 RNA
   12 Replicate
   13 Daughter cell
   14 Protein synthesis
   15 Messenger RNA
   16 Transcribe
   17 Ribosome
   18 Triplet
   19 Translate
   20 Precipitate out
   21 Chromosome
   22 Histone
   23 DNA profiling
   24 Extract
   25 Amplified DNA
   26 Restriction enzyme
   27 Gel electrophoresis
   28 DNA fingerprint
   29 Forensic
   30 Paternity case
   31 Genotype
   32 Variation
   33 Phenotype
   34 Gene expression
   35 Sexual reproduction
   36 Mutation
   37 Meiosis
   38 Gametes
   39 Mutagen
   40 Human Genome Project
   41 Genetic disorder
   42 Genetic screening
   43 Diagnostic test
   44 *Nucleic acid
   45 *Nucleotide
   46 *Nitrogenous base
   47 *Hydrogen bonds
   48 *Complementary strands
   49 *Antiparallel
   50 *Nucleosomes
   51 *Chromatin
   52 *Template
   53 *Semi-conservative
   54 *Peptide bond
   55 *Polypeptide chain

2. Distinguish between the following pairs of terms:
   (a) DNA and RNA
   (b) Mutation and mutagen
   (c) Maternal and paternal chromosomes
   (d) Chromosome and ribosome
   (e) Genetic screening and profiling
   (f) Chromosome mutation and gene mutation
   (g) Cystic fibrosis and Down's syndrome
   (h) *Deoxyribose and ribose
   (i) *Uracil and thymine
   (j) *mRNA and tRNA
   (k) *Purine and pyrimidine
   (l) *Transcription and translation
   (m) *Codon and anticodon

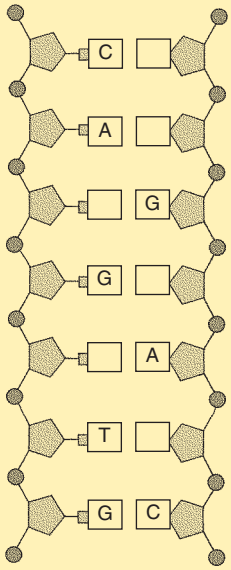
3. (a) Explain the meaning of the term genetic code.
   (b) Why is it necessary for DNA to be able to produce exact copies of itself?
   (c) Except for identical twins, everybody's DNA is unique. In what way is a person's DNA unique? Why are identical twins an exception?

4. Make a table comparing DNA with RNA.

5. DNA is mainly in the nucleus of a cell, but the enzymes and other proteins are made in the cytoplasm. So the message in the DNA has to get from the nucleus to the cytoplasm. Explain how this happens.

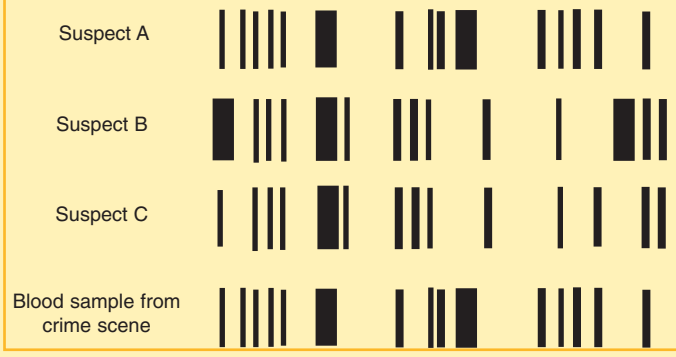
6. (a) The concentration of DNA per cell nucleus was found to be approximately the same in the kidney, heart and pancreas of an animal. Using your knowledge of DNA and mitosis suggest why the DNA concentration is approximately the same in these different cell types.
   (b) Referring to DNA, explain why a gardener uses vegetative propagation instead of the seed to propagate a particular variety of a plant.

7. The diagram shows a partly completed section of a DNA molecule.



(a) Copy and complete the diagram by writing the letters of the missing bases in the boxes.
(b) Mutations can occur because of changes to DNA.
(i) Name two possible causes for an increase in the normal rate of mutation.
(ii) Explain how a change in the DNA structure can cause a cell to form a different protein.

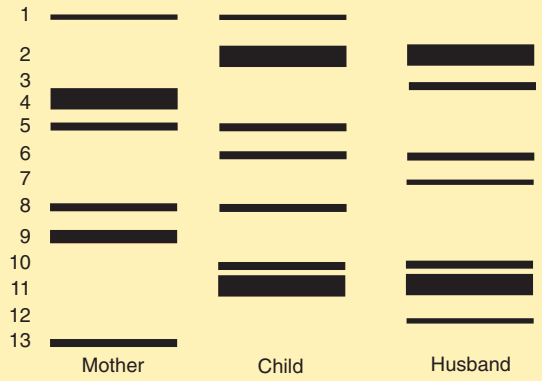
8. (a) Why is the term DNA profiling better than DNA fingerprinting?
(b) Do you think suspects should be convicted on the basis of their DNA profiles?
(c) In what circumstances might it be necessary to find out if two people are related?
(d) Some people feel that DNA profiling is an invasion of people's privacy. What do you think? In what ways might the technique be misused?
(e) The picture shows the DNA banding patterns of three suspects and a bloodstain found on the handle of a murder weapon. Which suspect is most likely to be guilty? Give a reason for your answer.



(f) Describe the stages involved in obtaining the DNA profile shown.

9. Genes control the formation of proteins. The DNA base sequence of the gene is translated into the amino acid sequence of the protein.
(a) Describe, with the aid of a labelled diagram, how a protein is synthesised in a cell.
(b) Give one reason why protein formation does not take place inside the nucleus.

10. It is sometimes necessary for members of a family to prove that they are related. One way in which scientists can help is by DNA fingerprinting.
(a) What is the function of DNA in a cell?
(b) In a DNA fingerprinting test cells were taken from the mother, the husband and the child. The cells were broken open and DNA fragments were extracted. The DNA fragments in each sample were separated into a column of bands, called a DNA fingerprint. The diagram shows the DNA fingerprint of each member of the family. The bands have been numbered to help you answer the questions which follow.



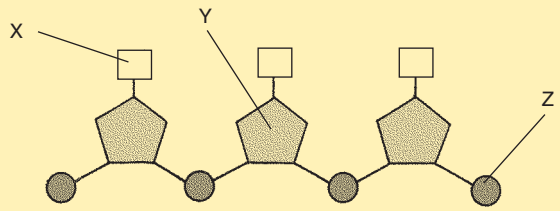
(i) Which of the bands in the child's DNA fingerprint are in exactly the same position as the bands in the mother's DNA fingerprint?
(ii) Explain why some of the bands in the mother's DNA fingerprint are absent from the child's DNA fingerprint.
(iii) The bands in the child's DNA fingerprint which are absent from the mother's DNA fingerprint must have been inherited from the child's natural father. Does the husband's DNA fingerprint show that he is the child's father? Explain your answer.

11. Give a scientific explanation for each of the following steps taken in the activity to separate DNA from a plant tissue extract.
    (a) Chop/blend the tissue for a few seconds.
    (b) Add the chopped tissue to a salt solution containing washing-up liquid.
    (c) Stand the container of chopped tissue in a water bath at 60°C for 10 minutes.
    (d) Cool the container of chopped tissue for a few minutes.
    (e) Filter the cooled mixture.
    (f) Add 2 to 3 drops of protease enzyme to the filtrate.
    (g) Add ice-cold alcohol to the surface of the plant tissue extract.
    (h) Dip a wire loop into the alcohol.

12. (a) What is meant by DNA profiling?
    (b) Describe briefly how DNA profiling is carried out.
    (c) Give two uses of DNA profiling.
    (LC OL Exam Paper)

HIGHER LEVEL

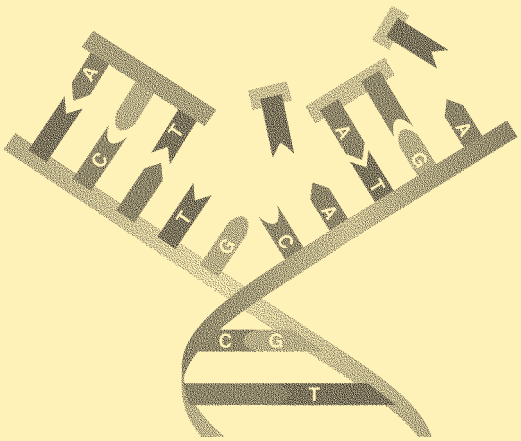
13. The diagram shows part of a polynucleotide chain from a DNA molecule.



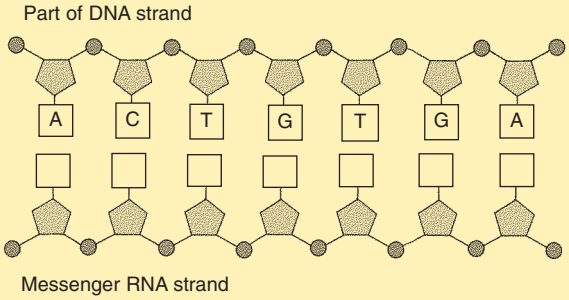
    (a) Copy the diagram and put a circle around one nucleotide.
    (b) Name the chemical groups labelled X, Y and Z.
    (c) How are the bases on the two adjacent strands of the DNA molecule held together?
    (d) Which base pairs up with cytosine in this molecule?
    (e) One of the bases in DNA is not represented in RNA. Name the base not represented. Name the base that replaces it in RNA.
    (f) Besides the nucleus, name one other place in the cell where DNA is located.

14. The sequence of nucleotides down the length of DNA forms the genetic code. Each sequence of three nucleotides, for example CGA or CCA, codes for one amino acid.
    (a) How many different groups of two letters can you make using C, G, A and T? (Each letter can be used once, twice or left out altogether.)
    (b) How many different groups can you make using three letters?

15. Nucleic acids are long-chain molecules consisting of units called nucleotides each containing one of five different bases denoted by the letters: A, C, G, T and U. The diagram shows a section of a DNA molecule beginning to 'unzip'. Letters representing some of the bases are missing from parts of the drawing.



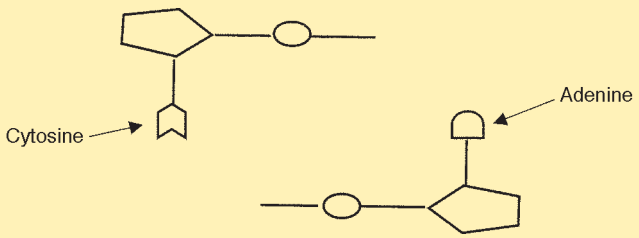
    (a) Write in the letters of the missing complementary bases.
    (b) A DNA molecule unzips and replicates itself just before mitosis. Why does it need to replicate?
    (c) When unzipped, a strand of DNA can act as a template for building molecules of messenger RNA. Copy and complete the diagram by writing in the letters of the missing complementary bases.



(d) The sequence of bases in a molecule of messenger RNA determines the sequence of amino acids in a polypeptide chain.
(i) How many bases are necessary to code for each amino acid in a polypeptide chain?
(ii) Where in the cell are amino acids assembled into polypeptides?
(iii) Explain why a change in the sequence of bases in the DNA is likely to change the structure of the protein made by the cell.

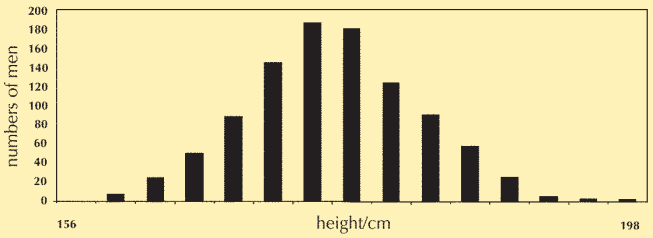
16. Answer the following in relation to protein synthesis.
(a) Outline the main events of transcription.
(b) What is a triplet? State the role of a triplet.
(c) Describe the role of tRNA.
(LC HL Sample Paper)

17. (a) Copy the diagram into your answer book and then complete it to show the complementary base pairs of the DNA molecule. Label all parts not already labelled.



(b) The genetic code incorporated into the DNA molecule finds its expression in part in the formation of protein. This formation requires the involvement of a number of RNA molecules. List these RNA molecules and briefly describe the role of each of them.
(LC HL Sample Paper)

18. The diagram shows the distribution of heights in a group of men between the ages of 18 and 23.



(a) What term is used by biologists to describe differences within a population with respect to features such as height?
(b) State two factors that could be responsible for the differences shown.
(c) Would you expect a similar distribution if the students were weighed instead of being measured for height? Explain your answer.
(d) What is a mutation?
(e) State one cause of mutation.
(f) Give an example of a condition, found in the human population, that results from a mutation.
(LC HL Exam Paper)